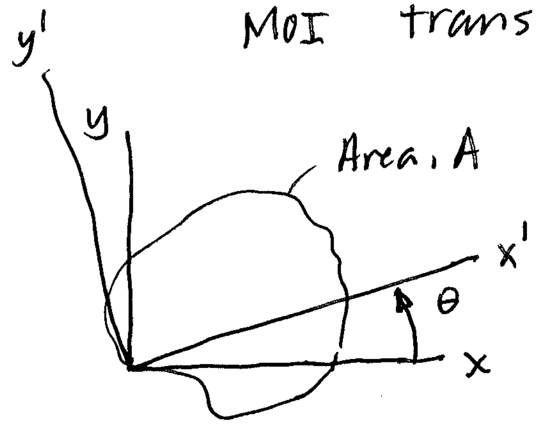
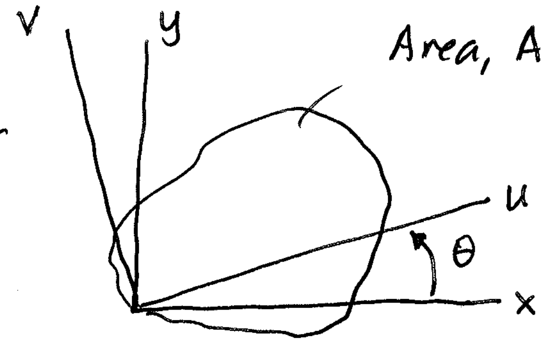
Derivation of Mohr's Circle

Earlier we derived the 2θ forms of the MOI transformation eqns.



or



Given for an area, A: I_x, I_y, I_{xy}

Find $\underbrace{I_{x'},\ I_{y'},\ I_{x'y'}}_{\text{Book}}$ or $\underbrace{I_u,\ I_v,\ I_{uv}}_{\text{I prefer.}}$ at θ.

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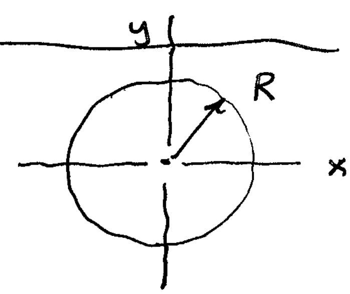
① $$I_{x'} = I_u = \left(\frac{I_x + I_y}{2}\right) + \left(\frac{I_x - I_y}{2}\right)\cos 2\theta - I_{xy}\sin 2\theta$$

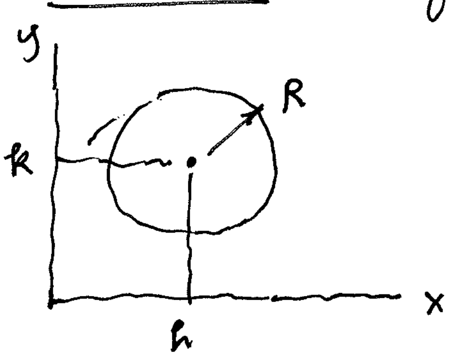
② $$I_{x'y'} = I_{uv} = I_{xy}\cos 2\theta + \left(\frac{I_x - I_y}{2}\right)\sin 2\theta$$

Recall:

Eqn for a general circle centered at origin:

$$\boxed{x^2 + y^2 = R^2}$$



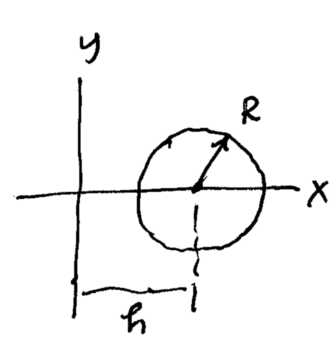


Circle w/ center (h, k):

$$\boxed{(x-h)^2 + (y-k)^2 = R^2}$$

Mohr's Circle is of this form:

$$\boxed{(x-h)^2 + y^2 = R^2}$$



From 2θ equations

let $\left(\frac{I_x + I_y}{2}\right) = I_{AVG}$

From ①: $I_u - I_{AVG} = \overbrace{\left(\frac{I_x - I_y}{2}\right)}^{= A} \cos 2\theta - I_{xy} \sin 2\theta$

② $I_{uv} = \overbrace{\left(\frac{I_x - I_y}{2}\right)}^{= A} \sin 2\theta + I_{xy} \cos 2\theta$

Square both sides and add:

$$(I_u - I_{AVG})^2 + I_{uv}^2$$

Cancel

$$= A^2 \cos^2 2\theta + I_{xy}^2 \sin^2 2\theta - \cancel{2A I_{xy} \cos 2\theta \sin 2\theta}$$
$$+ A^2 \sin^2 2\theta + I_{xy}^2 \cos^2 2\theta + \cancel{2A I_{xy} \cos 2\theta \sin 2\theta}$$

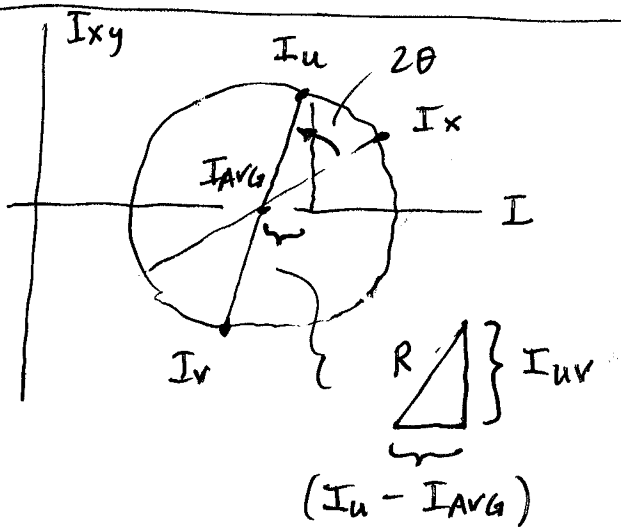
$$= A^2 \underbrace{(\cos^2 2\theta + \sin^2 2\theta)}_{1}$$
$$+ I_{xy}^2 \underbrace{(\cos^2 2\theta + \sin^2 2\theta)}_{1}$$

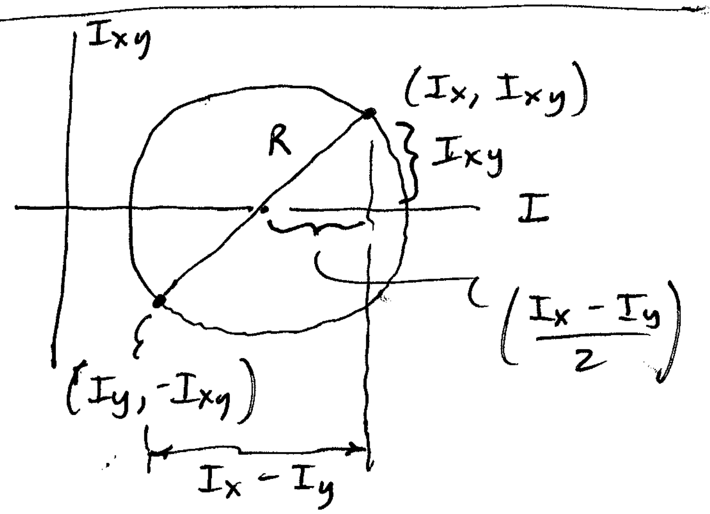
End up with:

$$\left((I_u - I_{AVG})^2 + I_{uv}^2\right) = \left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2$$
$$= R^2$$

LHS are I_u, I_{uv} at various θ's

RHS and thus R are invariant, based on known I_x, I_y, I_{xy}

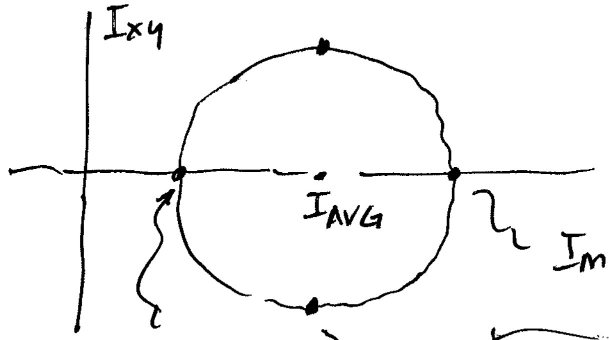




So the MC is set up with I_x, I_y, I_{xy}:

- Find center: $\boxed{I_{AVG} = \left(\frac{I_x + I_y}{2}\right)}$

- Find Radius: $R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$



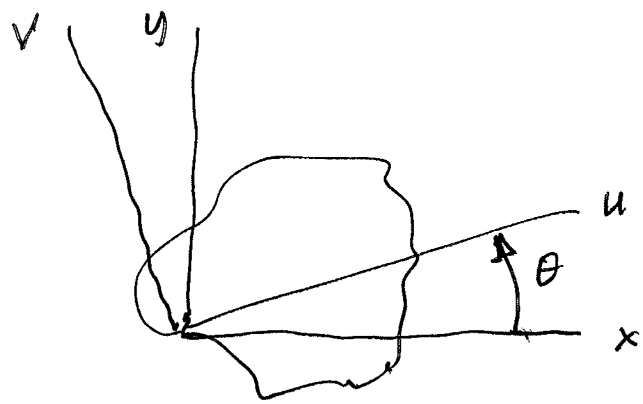
$I_{MAX} = I_{AVG} + R$

$I_{MIN} = I_{AVG} - R$

★ Note: at I_{MAX} + I_{MIN}, $I_{xy} = 0$.

At $I_x = I_y = I_{AVG}$, then I_{xy} = max value $= R$

Mohr's circle (MC) is the locus of all possible combinations of (I_u, I_{uv}), $(I_v, -I_{uv})$ at all possible θ's rotation from ${}^{y}L_{x}$.



2θ on MC corresponds to θ on original element

How find angle θ_p from x to I_{MAX} or I_{MIN} ?

Consider I_{uv} equation:

$$\boxed{I_{uv} = \left(\frac{I_x - I_y}{2}\right)\sin 2\theta + I_{xy}\cos 2\theta}$$

From MC, note that $I_{uv} = 0$ at I_{MAX}, I_{MIN}.

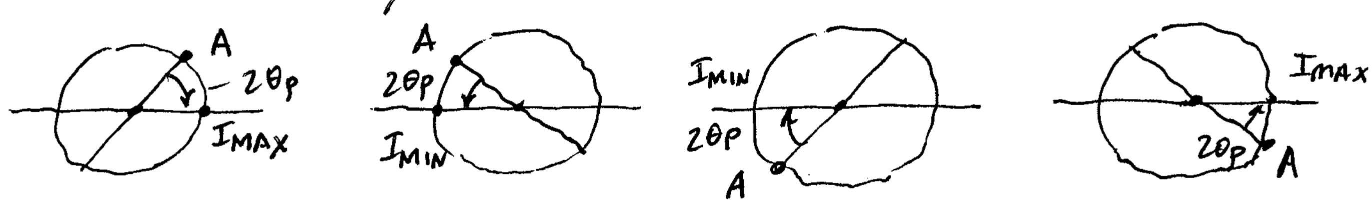
$$0 = \left(\frac{I_x - I_y}{2}\right)\sin 2\theta_p + I_{xy}\cos 2\theta_p$$

$$\boxed{\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}}$$

or

$$\boxed{\theta_p = \frac{1}{2}\tan^{-1}\frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}}$$

Once plot circle from I_{AVG} and R, then wish to locate point $A\,(I_x, I_{xy})$ on the circle. It can be anywhere on the circle:



$2\theta_p$ from $A\,(I_x, I_{xy})$ takes you to the nearest principal MOI, either I_{MAX} or I_{MIN}.